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Descriptive Analytics of U.S. Electric Grid Disruptions: Trends, Regional Patterns, and Operational Insights

This project applies descriptive analytics techniques using Python to analyze electric grid disruption data from 2000 to 2014. The dataset includes information such as event descriptions, dates, geographic regions, and the number of customers affected. The purpose of this analysis is to identify patterns and trends in grid disruptions and better understand factors influencing grid reliability.

This analysis is also relevant to the regional grid structure in the El Paso and Las Cruces area. While most of Texas operates under ERCOT, El Paso Electric is part of the WECC region. This distinction highlights the importance of understanding regional differences in grid operations, as system behavior, planning, and reliability considerations can vary significantly across governing interconnections.

Data Acquisition and Preparation

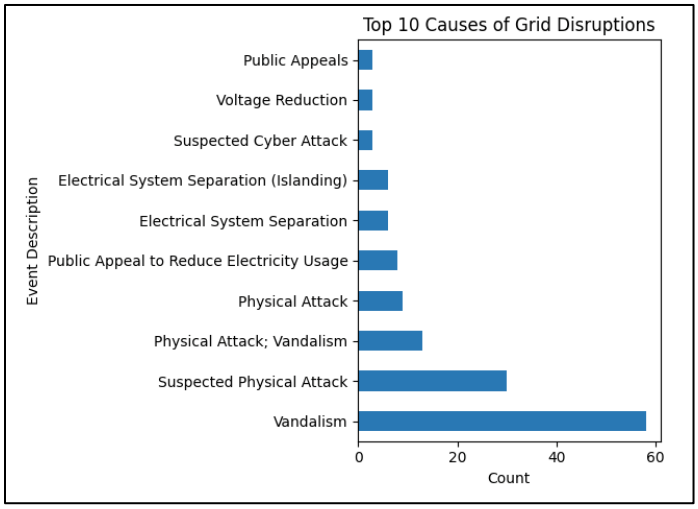
The dataset used in this analysis was obtained from Kaggle and contains information on electric grid disruptions from 2000 to 2014. The data includes variables such as event description, date, geographic region, and number of customers affected.

Data preparation involved cleaning and transforming the dataset to ensure accuracy and usability. Column names were standardized, date fields were converted to datetime format, and numerical variables such as demand loss and the number of customers affected were converted from text to numeric values. Missing values were handled by removing incomplete records for key variables to ensure reliable analysis.

Exploratory Data Analysis (EDA) was conducted to understand the dataset's structure and characteristics. Summary statistics were generated to evaluate the distributions of numerical variables, such as demand loss and the number of customers affected. Missing values were identified and addressed, and the dataset was reviewed for consistency and potential duplicates.

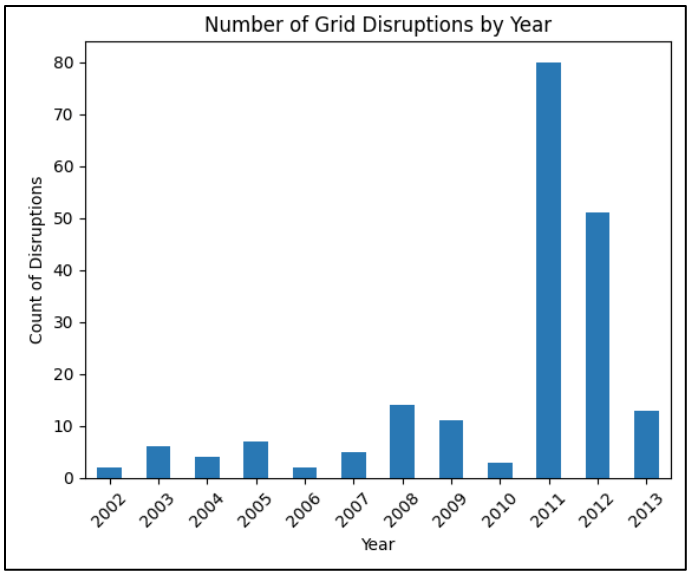
The analysis focused on identifying patterns in disruption causes, frequency over time, regional differences, and the impact on customers. Distributions and relationships among variables were explored through visualizations, enabling the identification of trends, variability, and outliers.

Figure 1: Top Causes of Grid Disruptions



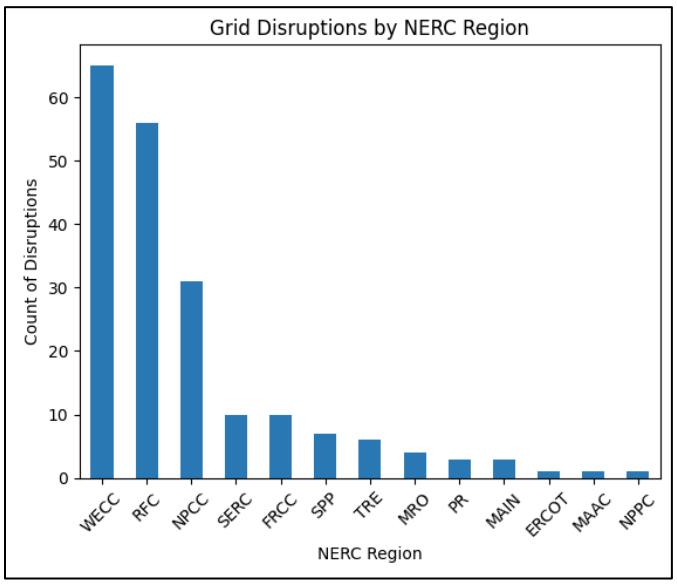
The chart illustrates the most common causes of grid disruptions. The results indicate that vandalism and physical attacks are among the most frequent causes, along with operational and system-related issues. This suggests that grid reliability is influenced not only by environmental factors but also by human-related incidents. Understanding these causes is important for improving system security, maintenance practices, and overall grid resilience.

Figure 2: Number of Grid Disruptions by Year



The chart shows that the number of grid disruptions varies significantly across the years. From 2002 to 2010, the number of events remained relatively low and stable. However, there is a sharp increase in disruptions around 2011, followed by a slightly lower but still elevated level in 2012. This spike may be due to changes in reporting practices, major weather events, or system-related factors during that period. Overall, the data indicate variability rather than a consistent trend over time.

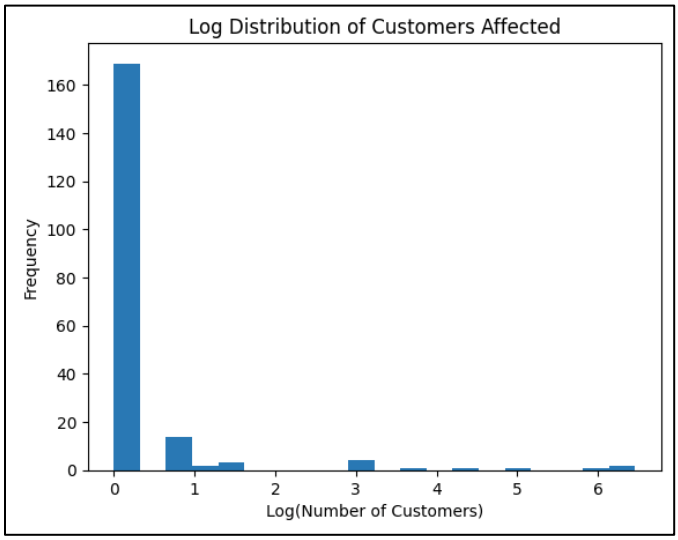
Figure 3: Grid Disruptions by NERC Region



The chart shows that grid disruptions are not evenly distributed across NERC regions. Regions such as WECC and RFC report the highest number of disruption events, while other regions experience significantly fewer occurrences. This variation may be influenced by factors such as geographic size, differences in infrastructure, and environmental exposure.

This finding is particularly relevant to the El Paso and Las Cruces area, where El Paso Electric operates within the WECC region, even though much of Texas is part of ERCOT. This regional distinction is important for system operations, as planning, reliability standards, and response strategies can differ between interconnections. Understanding these differences provides a better context for analyzing grid performance at a local level.

Figure 4: Distribution of Customers Affected (Log Scale)



The log-transformed histogram provides a clearer view of the distribution of customers affected by grid disruptions. While the original data is heavily skewed toward smaller values, applying a logarithmic transformation reveals a broader spread across the dataset. The majority of events still impact a relatively small number of customers, but the transformation makes it easier to observe the presence and scale of larger disruption events. This approach improves interpretability and highlights the variability in outage impact.

Future Research

Future analysis could expand this study by incorporating additional variables such as weather conditions, infrastructure age, or system load data. Including more recent data would improve understanding of long-term trends. Additionally, predictive modeling techniques could be applied to forecast disruptions and improve proactive grid management.

Conclusion

This project demonstrates how descriptive analytics can be used to analyze data on electric grid disruptions and identify meaningful patterns. The analysis highlights variability in disruption frequency, regional differences, and the presence of high-impact events affecting large numbers of customers. These findings emphasize the importance of data-driven decision-making to improve grid reliability and support effective system operations. These insights are especially relevant for regions such as El Paso and Las Cruces, where understanding the WECC interconnection is critical for effective planning and reliability management.

Sources:

Armstrong, A. (n.d.). *15 Years of Power Outages*. Kaggle. Retrieved from https://www.kaggle.com/datasets/autunno/15-years-of-power-outages